PROJECT REPORT

Academic Project — 2025-26

Course Management System

Built with Java Swing | Object Oriented Programming

Subject	Computer Science / Information Technology
Language	Java (JDK 21)
Tools Used	VS Code, Java Swing, JDK 21
Academic Year	2025 - 2026

1. Introduction

The Course Management System (CMS) is a desktop application developed using Java and the Java Swing GUI library. The project is built as part of the Computer Science curriculum to demonstrate practical knowledge of Object Oriented Programming (OOP) concepts including Encapsulation, Inheritance, Polymorphism, and Abstraction.

The application allows three types of users — Admin, Teacher, and Student — to log in and perform specific tasks related to course and assignment management. The entire application runs as a standalone desktop window without requiring any internet connection or external database.

1.1 Purpose of the Project

The main purpose of this project is to:

- Demonstrate the use of Java OOP concepts in a real-world application
- Build a functional GUI desktop application using Java Swing
- Implement role-based access control for Admin, Teacher, and Student
- Manage courses, users, and assignments through a user-friendly interface

1.2 Scope

The system covers the following functional areas:

- User authentication with role-based login
- Admin panel to manage users and courses
- Teacher panel to manage and post assignments
- Student panel to view courses and assignments

2. Objectives

The key objectives of this project are:

- To design and develop a Course Management System using Java
- To apply OOP concepts: Encapsulation, Inheritance, Polymorphism, and Abstraction
- To implement a graphical user interface using Java Swing (JFrame, JPanel, JTable, JButton, etc.)
- To provide role-based access for Admin, Teacher, and Student users
- To allow dynamic addition of users, courses, and assignments at runtime
- To demonstrate a clean, dark-themed, professional UI design
- To produce a deployable desktop application (.jar and .exe format)

3. System Analysis

3.1 Existing System

In most schools and colleges, course management is done manually. Teachers maintain paper registers for attendance, hand out assignment sheets physically, and grades are recorded in notebooks. This process is slow, error-prone, and difficult to manage for large numbers of students.

3.2 Proposed System

The proposed Course Management System is a digital desktop application that replaces manual processes. It provides:

- A secure login system with role-based access
- A centralised system to manage courses, users, and assignments
- Real-time data display using interactive tables
- A modern dark-themed graphical interface built with Java Swing

3.3 Feasibility Study

Feasibility Type	Analysis
Technical	Built with Java (JDK 21) and Java Swing — both free and open source. Runs on any computer with Java installed.
Operational	Simple login and navigation. Any student or teacher can use it without training.
Economic	Zero cost — Java, VS Code, and all tools used are completely free.
Time	Developed within 2 days as an academic project. Fully functional within the deadline.

4. System Design

4.1 System Architecture

The application follows a layered architecture with three main layers:

Layer	Description
Presentation Layer	Java Swing — LoginFrame, MainFrame, all JPanel pages
Business Logic Layer	OOP classes — User, Admin, Teacher, Student, Course, Assignment
Data Layer	DatabaseManager class — stores all data in ArrayList objects

4.2 OOP Concepts Used

OOP Concept	Class / Location	How it is Applied
Encapsulation	User, Course, Assignment	All fields are private; accessed through public getters only
Inheritance	Admin, Teacher, Student extend User	Common properties inherited from base class User
Polymorphism	MainFrame sidebar menu	Same login method shows different menus based on role
Abstraction	LoginFrame, DatabaseManager	Internal logic hidden; only necessary methods exposed

4.3 Class Descriptions

Class Name	Purpose
User	Base class for all users. Contains id, name, email, password, role with private fields and public getters.
Admin	Extends User. Represents admin users. Role is set to 'admin' automatically.
Teacher	Extends User. Represents teacher users. Role is set to 'teacher' automatically.
Student	Extends User. Represents student users. Role is set to 'student' automatically.
Course	Stores course details — id, courseName, teacherName, description.
Assignment	Stores assignment details — id, courseId, title, dueDate, description.
DatabaseManager	Manages all data using ArrayLists. Provides login(), addUser(), addCourse(), addAssignment() methods.

LoginFrame	Java Swing JFrame window for login screen. Validates credentials and opens MainFrame.
MainFrame	Main application window with sidebar navigation and role-based pages.

5. Module Description

5.1 Login Module

The login module is the entry point of the application. It displays a professional dark-themed login window with fields for Email and Password. Quick Fill buttons are provided for easy testing. The system validates credentials against the DatabaseManager and redirects users to their respective panels based on role.

5.2 Admin Module

The Admin panel provides full control over the system. It includes the following pages:

- Dashboard — Shows summary statistics (total users, courses, assignments, students) and a full user list
- Manage Users — Allows adding new users with name, email, password, and role selection
- Courses — Allows viewing all courses and adding new courses with name, teacher, and description
- Assignments — Allows viewing all assignments and posting new assignments to any course

5.3 Teacher Module

The Teacher panel allows teachers to manage their teaching activities:

- My Courses — Displays all available courses
- Assignments — Allows posting new assignments and viewing all existing assignments

5.4 Student Module

The Student panel provides a read-only view for students:

- All Courses — View all available courses with teacher and description
- Assignments — View all assignments with title, course, due date, and description

6. Implementation

6.1 Technology Stack

Component	Technology Used
Programming Language	Java (JDK 21)
GUI Framework	Java Swing (javax.swing, java.awt)
IDE / Editor	Visual Studio Code with Extension Pack for Java
Data Storage	In-memory ArrayList (no external database required)
Deployment	Exported as .jar file, convertible to .exe using Launch4j

6.2 Sample Login Credentials

Role	Email	Password
Admin	admin@cms.com	admin123
Teacher	teacher@cms.com	teacher123
Student	student@cms.com	student123

6.3 Sample Data

The system loads the following sample data on startup:

Course	Teacher	Description
Mathematics	Mr. Kumar	Algebra, Geometry
Computer Science	Mr. Kumar	Java, OOP, DSA
Physics	Mr. Kumar	Mechanics, Waves

7. Testing

The system was tested manually for all major use cases. The following test cases were performed:

No.	Test Case	Expected Output	Status	Result
1	Login with valid Admin credentials	Dashboard opens	Passed	PASS
2	Login with invalid credentials	Error message shown	Passed	PASS
3	Admin adds a new user	User appears in table	Passed	PASS
4	Admin adds a new course	Course appears in table	Passed	PASS
5	Teacher posts assignment	Assignment appears in list	Passed	PASS
6	Student views courses	All courses displayed	Passed	PASS
7	Student views assignments	All assignments displayed	Passed	PASS
8	Logout and re-login	Login screen returns	Passed	PASS
9	Add user with empty fields	Error message shown	Passed	PASS
10	Stats update after adding data	Numbers update on dashboard	Passed	PASS

8. Limitations and Future Scope

8.1 Current Limitations

- Data is stored in memory only — all data is lost when the application is closed
- No real database integration (MySQL, SQLite) in the current version
- No file upload support for assignment submissions
- No email notification system for assignment reminders
- Password is stored as plain text (not encrypted)

8.2 Future Scope

- Integrate MySQL database for permanent data storage
- Add password hashing using SHA-256 for security
- Implement file upload for student assignment submissions
- Add grade entry and report card generation
- Develop a web-based version using Java Spring Boot and React
- Add email notification system for deadlines and announcements
- Implement attendance tracking module

9. Conclusion

The Course Management System was successfully developed using Java and Java Swing as part of the academic project requirement. The application demonstrates a practical implementation of Object Oriented Programming concepts including Encapsulation, Inheritance, Polymorphism, and Abstraction.

The system provides a role-based login for Admin, Teacher, and Student users, each with their own dedicated panel and features. The graphical user interface is built entirely using Java Swing components and offers a clean, professional, dark-themed design.

All planned features were implemented successfully and tested. The project was completed within the given deadline and meets all the academic requirements specified. Future enhancements such as database integration and a web-based interface can be added to make this a production-ready system.

10. Bibliography / References

- Oracle Java Documentation — https://docs.oracle.com/javase/21/
- Java Swing Tutorial — https://docs.oracle.com/javase/tutorial/uiswing/
- GeeksForGeeks Java OOP Concepts — https://www.geeksforgeeks.org/object-oriented-programming-oops-concept-in-java/
- W3Schools Java Tutorial — https://www.w3schools.com/java/
- Visual Studio Code Java Guide — https://code.visualstudio.com/docs/java/java-tutorial